

Nginx Multi-Site Troubleshooting Guide

1. “nginx: command not found”

Nginx isn't installed or isn't in your PATH.

```
which nginx
sudo apt install nginx -y      # Ubuntu/Debian
sudo dnf install nginx -y      # CentOS/RHEL
```

2. Nginx fails to start / “Job for nginx.service failed”

Check the detailed error log:

```
sudo systemctl status nginx
sudo journalctl -xeu nginx
```

Common causes: - A config syntax error → run `sudo nginx -t` to find the exact line. - Port already in use (see #3 below).

3. “Address already in use” (port conflict)

Another process is already listening on 8081/8082.

```
sudo ss -tulpn | grep 8081
sudo ss -tulpn | grep 8082
```

Either stop the conflicting process, or change `listen 8081;` in your config to a free port.

4. Config test fails: `nginx -t` shows errors

```
sudo nginx -t
```

This prints the exact file and line number of the problem. Common mistakes: - Missing semicolon `;` at the end of a directive. - Missing closing brace `}`. - Duplicate `server_name`/listen combination across two enabled files.

Fix the file, then re-run `nginx -t` until it reports `syntax is ok` and test is successful.

5. Site not enabled (Ubuntu/Debian)

If you created a config in sites-available but forgot to symlink it:

```
ls -l /etc/nginx/sites-enabled/  
sudo ln -s /etc/nginx/sites-available/site1 /etc/nginx/sites-  
enabled/  
sudo systemctl reload nginx
```

6. 403 Forbidden

Usually a permissions or missing-index-file issue.

```
ls -l /var/www/site1/  
sudo chmod -R 755 /var/www/site1  
sudo chown -R www-data:www-data /var/www/site1 # Ubuntu/Debian  
sudo chown -R nginx:nginx /var/www/site1 # CentOS/RHEL
```

Also confirm index.html actually exists in that folder and index.html; is set in the config.

7. 404 Not Found

- Confirm root in the server block points to the correct directory.
 - Confirm the file requested actually exists there.
 - Run `sudo nginx -t` and `sudo systemctl reload nginx` after any config change — edits don't apply until reloaded.
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8. Can't access the site from another machine / browser

- **Firewall blocking the port:**

```
sudo ufw status # Ubuntu  
sudo firewall-cmd --list-ports # CentOS/RHEL
```

Open the port if missing (see README.md Step 5).

- **Cloud provider security group:** if this is a VM on AWS/GCP/Azure/DigitalOcean, you also need to allow the port in the cloud provider's firewall/security group, separate from the OS firewall.
 - **Wrong IP:** use the server's actual public/private IP, not localhost, when testing from another machine.
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9. SELinux blocking Nginx (CentOS/RHEL/Fedora only)

Symptom: config is valid, port is open, but you still can't connect, and `/var/log/audit/audit.log` shows denied entries.

```
sudo semanage port -a -t http_port_t -p tcp 8081
```

```
sudo semanage port -a -t http_port_t -p tcp 8082
sudo systemctl reload nginx
```

If semanage isn't found:

```
sudo dnf install policycoreutils-python-utils -y
```

10. Changes not showing up in the browser

- Did you run `sudo systemctl reload nginx` (or restart) after editing the config/files?
 - Hard-refresh the browser (Ctrl+Shift+R) or clear cache — static files are often cached.
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Quick Diagnostic Checklist

```
sudo nginx -t                                # 1. Config syntax OK?
sudo systemctl status nginx                  # 2. Service running?
sudo ss -tulpn | grep nginx                  # 3. Listening on expected
ports?
curl -I http://localhost:8081                # 4. Responding locally?
sudo ufw status / firewall-cmd --list-ports # 5. Firewall open?
ls -l /var/www/site1                        # 6. Files present &
permissions correct?
sudo journalctl -xeu nginx --no-pager | tail -50 # 7. Recent
errors?
```

If all seven checks pass and it still doesn't work externally, the issue is almost always the cloud provider's network/security group rather than Nginx itself.